

Claude Code Max Master Prompt

AI-Driven Multimodal Sentiment Analysis for Mental Health Monitoring on Social Media

You are going to build my complete Final Year B.Tech Research Project from scratch.

IMPORTANT: I have provided a project synopsis PDF. You **MUST** treat that PDF as the primary specification for this project.

PROJECT TITLE:

"AI-DRIVEN MULTIMODAL SENTIMENT ANALYSIS FOR MENTAL HEALTH MONITORING ON SOCIAL MEDIA"

The project is being developed as a B.Tech Computer Science and Engineering / Data Science research project.

1. FIRST: FULLY UNDERSTAND THE PROJECT SYNOPSIS

Before writing substantial code, carefully read the ENTIRE attached PDF.

Do not assume that this is simply a sentiment-analysis web application.

The actual research problem is:

Social-media posts may contain both text and images. Text-only systems can miss emotional information contained in images, while image-only systems can miss meaning expressed through text. The project therefore investigates whether combining textual and visual information can improve mental-health-related sentiment classification.

The synopsis explicitly requires comparison between:

A. TEXT-ONLY MODEL

B. IMAGE-ONLY MODEL

C. MULTIMODAL TEXT + IMAGE MODEL

The central research question is:

"Can an AI-driven multimodal framework combining textual and visual information improve sentiment-based mental health monitoring on social media compared with unimodal approaches?"

The project should therefore be designed as an EXPERIMENTAL AND COMPARATIVE RESEARCH PROJECT, not merely as an application.

2. IMPORTANT ACADEMIC BOUNDARY

This system is NOT a clinical diagnostic system.

It must NEVER claim that the model diagnoses depression, anxiety, or another mental illness.

Instead, frame it as:

- sentiment/emotion analysis
- mental-health-related pattern detection
- computational monitoring

- research support
- early indication / risk-oriented analysis only when justified by the dataset

The system is NOT a replacement for qualified mental-health professionals.

Maintain this distinction throughout code, UI, README, research paper, report, presentation, and model output.

3. FIRST TASK — REQUIREMENTS ANALYSIS

DO NOT immediately start coding.

First create PROJECT_REQUIREMENTS_ANALYSIS.md.

Extract:

- Project title
- Problem statement
- Background
- Motivation
- Primary research question
- Secondary research questions
- Research objectives
- Project scope
- Literature themes
- Research gap
- Proposed framework
- Required modalities
- Required comparison
- Dataset requirements
- Evaluation metrics
- Expected research contribution
- SDG relevance
- Expected challenges
- Timeline
- Team responsibilities
- Expected publication/research outcome

Separate MANDATORY REQUIREMENTS from OPTIONAL/EXTENSIBLE FEATURES.

4. RESEARCH-FIRST APPROACH

Before selecting technologies or datasets, conduct serious research on:

- Multimodal sentiment analysis
- Multimodal emotion recognition
- Mental-health-related social-media analysis
- Text-based mental health analysis
- Image-based emotion analysis
- Multimodal fusion

- Transformer-based NLP
- Vision Transformers / CNNs
- Cross-modal attention
- Multimodal transformers
- Explainable AI
- Social-media bias
- Multimodal dataset limitations

Prefer recent peer-reviewed research from reliable sources.

Do not fabricate papers or citations.

For each research paper used, record title, authors, year, venue, DOI/URL, methodology, dataset, results, limitations, and relevance.

Create LITERATURE_REVIEW.md.

5. IDENTIFY THE REAL RESEARCH GAP

Do NOT claim "no one has done this before."

Determine a defensible research gap from the literature, such as:

- limitations in multimodal fusion
- insufficient comparison between modalities
- conflicting textual and visual signals
- dataset limitations
- class imbalance
- explainability
- robustness
- cross-domain generalization
- multilingual limitations
- computational efficiency
- interpretability

Create RESEARCH_GAP.md answering:

1. What has already been done?

2. What are the limitations?

3. What remains insufficiently studied?

4. What will our project investigate?

5. Why is the proposed experiment meaningful?

6. AUTHENTIC DATASET — EXTREMELY IMPORTANT

Do NOT create synthetic/fake social-media posts and present them as real data.

Do NOT use a random Kaggle dataset merely because it is easy to download.

The dataset MUST contain:

- Text
- Corresponding image/visual content
- Appropriate labels
- Sufficient samples
- Legitimate provenance
- Research usability

Investigate credible academic sources and repositories.

Evaluate candidate datasets on:

- 1. Authenticity**
- 2. Provenance**
- 3. Original publication**
- 4. Number of samples**
- 5. Text availability**
- 6. Image availability**
- 7. Label quality**
- 8. Label semantics**
- 9. Class distribution**
- 10. Missing modalities**
- 11. Licensing**
- 12. Research reproducibility**
- 13. Ethical/privacy issues**
- 14. Suitability**
- 15. Computational feasibility**

Create DATASET_SELECTION.md with a comparison of serious candidates and a justification for the final choice.

If no single dataset perfectly satisfies every requirement, explain the limitation and choose the most scientifically defensible setup.

7. DATASET ETHICS AND PRIVACY

Social-media data can contain sensitive information.

Do NOT expose usernames, personal identifiers, private content, unnecessary URLs, or personally identifiable information.

Do not scrape private accounts.

Respect dataset licensing.

Create ETHICS_AND_PRIVACY.md covering privacy, consent where known, licensing, potential harm, bias, representation, false positives, false negatives, misuse, stigma, and responsible deployment.

8. DEFINE LABELS CORRECTLY

Inspect actual dataset labels before training.

Do NOT arbitrarily rename labels.

Document exactly what each class means.

Do not claim ordinary sentiment labels are clinical diagnoses.

Model outputs must correspond exactly to the dataset's actual target labels.

9. PROJECT ARCHITECTURE

Create PROJECT_ARCHITECTURE.md.

Design:

TEXT PIPELINE:

Raw Text -> Cleaning -> Tokenization -> NLP Representation -> Text Embedding

IMAGE PIPELINE:

Image -> Resize/Normalize -> CNN or Vision Transformer -> Visual Embedding

MULTIMODAL PIPELINE:

Text Embedding + Image Embedding -> Fusion -> Classification Head -> Output

Do not automatically use simple concatenation. Research and evaluate early fusion, late fusion, feature concatenation, weighted fusion, attention-based fusion, cross-modal attention, or multimodal transformers as appropriate.

10. REQUIRED EXPERIMENTAL DESIGN

Implement at least:

MODEL A — TEXT ONLY

MODEL B — IMAGE ONLY

MODEL C — MULTIMODAL

The comparison is the core research experiment.

Do not assume the multimodal model will win. If it performs worse, report and investigate why.

11. STRONGER RESEARCH EXPERIMENTS

Where feasible, perform:

- traditional/simple NLP baseline
- transformer-based text model
- CNN/vision baseline
- modern image model
- simple multimodal fusion
- improved multimodal fusion
- ablation study
- error analysis
- class imbalance analysis
- explainability analysis

Every experiment must answer a research question.

12. ABLATION STUDY

Perform a meaningful ablation study such as:

Text only

Image only

Text + Image

And, where appropriate, compare different fusion strategies.

The exact design must depend on the chosen architecture.

13. EVALUATION

Do not rely only on accuracy.

Use appropriate metrics:

- Accuracy
- Precision
- Recall
- F1-score
- Macro F1
- Weighted F1
- Confusion Matrix
- ROC-AUC where appropriate
- PR-AUC where appropriate

For imbalanced classes, prioritize macro-F1 and class-wise metrics.

Analyze per-class performance, false positives, false negatives, and confusion patterns.

Every reported number MUST come from actual experiments.

14. STATISTICAL / RESEARCH VALIDITY

Where practical, use:

- cross-validation
- confidence intervals
- repeated experiments
- statistical comparison
- significance testing where appropriate
- variance across runs

Do not apply statistical tests blindly. Explain why each method is appropriate.

15. MULTIMODAL CONFLICT ANALYSIS

Investigate posts where text suggests one emotional state but the image suggests another, if the dataset permits.

Analyze whether multimodal models handle conflicting signals better or worse.

Create MODALITY_CONFLICT_ANALYSIS.md if feasible.

16. EXPLAINABLE AI

Where appropriate implement:

Text: attention visualization, SHAP, LIME, or token importance.

Images: Grad-CAM, saliency maps, or attention visualization.

Multimodal: investigate how each modality contributed.

Do not create misleading explanations. Document limitations.

17. MODEL DEVELOPMENT

Consider suitable pretrained models such as:

Text: BERT, RoBERTa, DistilBERT, DeBERTa, or another justified transformer.

Image: ResNet, EfficientNet, ConvNeXt, Vision Transformer, CLIP/image encoder.

Multimodal: feature fusion, attention fusion, cross-modal attention, multimodal transformer.

Do NOT automatically choose the largest model.

Choose based on dataset size, GPU availability, performance, training cost, reproducibility, and research value.

18. COMPUTATIONAL CONSTRAINTS

The project must actually be executable.

Optimize for Google Colab / available NVIDIA GPU / normal student environments.

Use transfer learning where appropriate and provide CPU-compatible fallback where feasible.

19. DATA LEAKAGE PREVENTION

Ensure:

- correct train/validation/test split
- no preprocessing leakage
- no duplicate posts/images across splits
- correct augmentation
- no test-set use during tuning

Document the leakage-prevention strategy.

20. REPRODUCIBILITY

Use:

- fixed random seeds where appropriate
- requirements.txt or pyproject.toml
- configuration files
- versioned experiment settings
- saved model configurations
- dataset documentation
- experiment logs

Create REPRODUCIBILITY.md.

21. PROJECT STRUCTURE

Create a clean repository such as:

project/

```

■■■■ README.md
■■■■ PROJECT_REQUIREMENTS_ANALYSIS.md
■■■■ RESEARCH_GAP.md
■■■■ LITERATURE_REVIEW.md
■■■■ DATASET_SELECTION.md
■■■■ ETHICS_AND_PRIVACY.md
■■■■ PROJECT_ARCHITECTURE.md
■■■■ REPRODUCIBILITY.md
■■■■ REFERENCES.md
■■■■ ORIGINALITY.md
■■■■ data/
■■■■ notebooks/
■■■■ src/
■ ■■■ data/
■ ■■■ preprocessing/
■ ■■■ text_model/
■ ■■■ image_model/
■ ■■■ multimodal/
■ ■■■ evaluation/
■ ■■■ explainability/
■ ■■■ utils/
■■■■ experiments/
```



```
■■■ models/
■■■ results/
■■■ visualizations/
■■■ app/
■■■ tests/
■■■ docs/
■■■ research_paper/
```

Modify the structure if needed.

22. BUILD THE COMPLETE APPLICATION

After the research pipeline works, build a usable demonstration application allowing:

- Text input
- Image input
- text-only prediction
- image-only prediction
- multimodal prediction
- confidence/probability
- comparison between modalities
- relevant explanations/visualizations

Clearly display that the system is for research/monitoring support and is not a clinical diagnostic tool.

23. DASHBOARD

Build a research dashboard showing:

- dataset statistics
- class distribution
- model comparison
- accuracy
- precision
- recall

- F1

- confusion matrices
- ROC/PR curves where applicable
- model information
- multimodal improvement
- error analysis
- explainability outputs

24. TESTING

Create tests for:

- preprocessing
- dataset loading

- text pipeline
- image pipeline
- multimodal fusion
- prediction

- API

- UI/backend integration where practical
- edge cases

Actually run the tests.

Create TESTING.md.

25. SECURITY

Check:

- hardcoded secrets
- unsafe file uploads
- malicious input
- path traversal
- dependency vulnerabilities
- unsafe API endpoints
- sensitive information leakage

Use environment variables and create .env.example.

26. ORIGINALITY

The goal is genuine independent academic work, not bypassing plagiarism detection.

Do NOT copy GitHub repositories, Kaggle notebooks, research-paper code, tutorials, project reports, or architectures word-for-word.

Libraries and pretrained models are allowed and must be acknowledged.

Create ORIGINALITY.md documenting:

- independently developed components
- research inspiration
- pretrained models
- libraries
- experimental contribution
- engineering contribution
- research contribution

Never make an absolute claim of global uniqueness.

27. RESEARCH CONTRIBUTION

Aim for defensible contributions such as:

- 1. Systematic comparison of text-only, image-only and multimodal approaches.**
- 2. Investigation of multimodal performance improvement.**
- 3. Analysis of conflicting text/image emotional signals.**
- 4. Evaluation of fusion mechanisms.**
- 5. Explainability of multimodal predictions.**
- 6. Robust evaluation under class imbalance.**
- 7. Analysis of modality-specific failures.**

The final contribution must depend on actual experimental evidence.

28. RESEARCH PAPER

Only AFTER experiments are completed, create:

research_paper/

■■■ paper.md

■■■ paper.tex

■■■ references.bib

■■■ figures/

Include:

- Title
- Abstract
- Keywords
- Introduction
- Related Work
- Research Gap
- Research Questions
- Dataset
- Methodology
- Text-only model
- Image-only model
- Multimodal model
- Feature fusion
- Experimental setup
- Results
- Model comparison
- Ablation study
- Error analysis
- Explainability
- Discussion
- Ethics

- Limitations
- Future work
- Conclusion
- References

Every result must come from actual code.

Every citation must be real.

Every table and figure must use actual results.

29. FINAL YEAR PROJECT REPORT

Create a complete report containing:

- Abstract
- Introduction
- Motivation
- Problem Statement
- Objectives
- Literature Review
- Research Gap
- Existing System
- Proposed System
- Requirements
- System Architecture
- Methodology
- Dataset
- Preprocessing
- Model Design
- Multimodal Fusion
- Implementation
- Experiments
- Results
- Testing
- Ethics
- Societal Impact
- SDG Alignment
- Limitations
- Future Scope
- Conclusion
- References

The synopsis identifies:

Primary SDG: SDG 3 — Good Health and Well-being

Secondary: SDG 9 — Industry, Innovation and Infrastructure

Secondary: SDG 10 — Reduced Inequalities

Maintain the intended alignment without exaggerating impact.

30. VIVA PREPARATION

Create VIVA_PREPARATION.md with questions and answers about:

- project motivation
- multimodal approach
- dataset
- research gap
- novelty
- NLP model
- image model
- fusion method

- BERT

- CNN

- transfer learning
- F1-score
- imbalance
- overfitting
- data leakage
- false positives/negatives
- ethics
- clinical diagnosis limitation
- limitations
- future work
- research contribution
- cases where multimodal performs worse

Answers should be technically correct but student-friendly.

31. PRESENTATION

Create PRESENTATION_CONTENT.md with about 15–20 slides:

1. Title

2. Problem

3. Motivation

4. Existing Approaches

5. Research Gap

6. Objectives

7. Dataset

8. Proposed Architecture

9. Text Pipeline

10. Image Pipeline

11. Multimodal Fusion

12. Experimental Setup

13. Results

14. Model Comparison

15. Explainability

16. Application/Dashboard

17. Research Contribution

18. Limitations

19. Future Work

20. Conclusion

32. FINAL AUDIT

Create FINAL_PROJECT_AUDIT.md and verify:

- ☐ Synopsis requirements understood
- ☐ Text-only model implemented
- ☐ Image-only model implemented
- ☐ Multimodal model implemented
- ☐ Authentic dataset selected
- ☐ Dataset provenance documented
- ☐ Dataset labels verified
- ☐ Dataset licensing checked
- ☐ Ethics documented
- ☐ Data leakage checked
- ☐ Experiments executed
- ☐ Results generated from real data
- ☐ Metrics calculated correctly
- ☐ Ablation completed where feasible
- ☐ Error analysis completed
- ☐ Explainability implemented where feasible
- ☐ Application functional
- ☐ Tests executed
- ☐ Documentation complete
- ☐ Research paper generated

- ☐ Project report generated
- ☐ Viva preparation generated
- ☐ Presentation content generated
- ☐ No fabricated results
- ☐ No fabricated citations
- ☐ No fake dataset
- ☐ No hardcoded credentials

33. DEVELOPMENT BEHAVIOUR

RULE 1: Do not rush.

RULE 2: Do not write hundreds of files before validating architecture.

RULE 3: Actually execute important code.

RULE 4: Fix errors instead of hiding them.

RULE 5: Never fabricate experimental results.

RULE 6: Never fabricate citations.

RULE 7: Never fabricate dataset information.

RULE 8: Never claim clinical diagnosis.

RULE 9: Never copy existing projects.

RULE 10: Document important decisions.

RULE 11: Use current and appropriate research.

RULE 12: Prefer reproducibility over unnecessary complexity.

RULE 13: If a technique cannot realistically run in the environment, select a scientifically reasonable alternative and document why.

RULE 14: Write the research paper from actual implementation and results, not assumptions.

RULE 15: If an experiment performs poorly, keep and analyze the result rather than manipulating it.

34. EXECUTION ORDER

PHASE 1 — Read and understand synopsis

PHASE 2 — Requirements analysis

PHASE 3 — Literature research

PHASE 4 — Research-gap identification

PHASE 5 — Dataset investigation

PHASE 6 — Dataset selection and validation

PHASE 7 — System architecture

PHASE 8 — Data pipeline

PHASE 9 — Text-only baseline

PHASE 10 — Image-only baseline

PHASE 11 — Multimodal model

PHASE 12 — Fusion experiments

PHASE 13 — Evaluation

PHASE 14 — Ablation/error analysis

PHASE 15 — Explainability

PHASE 16 — Application/dashboard

PHASE 17 — Testing

PHASE 18 — Final experiments

PHASE 19 — Research results

PHASE 20 — Research paper

PHASE 21 — Final-year project report

PHASE 22 — Presentation

PHASE 23 — Viva preparation

PHASE 24 — Final audit

35. MOST IMPORTANT FINAL INSTRUCTION

Build this as a REAL RESEARCH PROJECT.

Do not optimize for lots of code or lots of features.

Optimize for:

- scientifically valid methodology
- authentic data
- reproducibility
- meaningful experiments
- defensible research gap
- measurable contribution
- proper multimodal comparison
- high-quality engineering
- responsible AI
- strong documentation
- genuine originality

The central experiment must remain:

TEXT ONLY

vs

IMAGE ONLY

vs

TEXT + IMAGE

The project should ultimately provide evidence-based answers to:

- 1. How well can textual information identify the target emotional/mental-health-related categories?**
- 2. What additional information does visual content provide?**
- 3. Does combining text and images improve classification?**
- 4. Under what conditions does multimodal fusion help?**

- 5. Under what conditions does it fail?**
- 6. Which fusion strategy works best for the selected dataset?**
- 7. Can the model's decisions be meaningfully explained?**
- 8. What limitations prevent this system from being considered a clinical diagnostic tool?**

START NOW.

First read the attached project synopsis completely.

Then create the requirements analysis and research plan BEFORE beginning full implementation.

Do not skip the research and dataset-selection stages.